

LAND INEQUALITY IN THE DEVELOPING WORLD

Luis Bauluz*, Yajna Govind†, Filip Novokmet‡, Daniel Sánchez Ordóñez§

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Abstract

Agricultural land is vital for three out of every four of the poorest billion individuals in the world, yet little is known about its distribution. Existing cross-country estimates of land inequality, based on agricultural census data, measure the size distribution of agricultural holdings. In practice, these neither reflect land ownership inequality nor value inequality and often do not account for the landless population. In this paper, we tackle these issues and provide novel and consistent estimates of land inequality across countries, based on household surveys. We show that (i) inequality in land value can differ significantly from inequality in land area, (ii) the proportion of landless households across countries varies substantially, markedly affecting estimates of inequality, and (iii) regional patterns in inequality according to our benchmark metric (land value inequality including the landless) contradict existing estimates from agricultural censuses. Overall, South Asia and Latin America exhibit the highest levels of inequality, with the top 10% of landowners capturing up to 75% of agricultural land, followed by Africa and “Communist” Asia (China and Vietnam) at levels of around 55–60%.

JEL classification: Q15, O1

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*University of Bonn, and World Inequality Lab

†Copenhagen Business School, and World Inequality Lab

‡University of Bonn, and World Inequality Lab

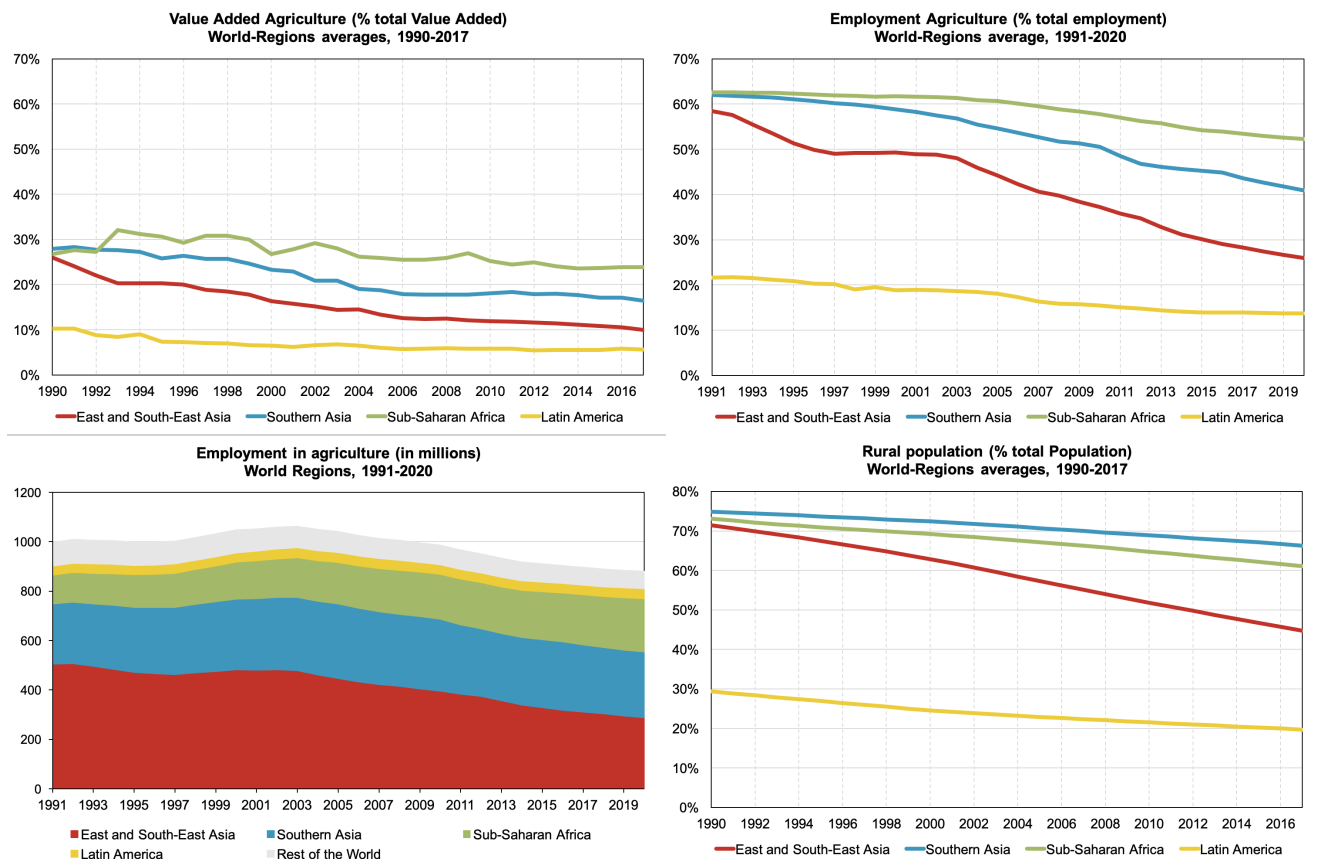
§Paris School of Economics

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1. Introduction

Agricultural land is vital for three out of every four of the poorest billion individuals in the world who depend on land and related activities for their subsistence (FAO, 2016). Over the past three decades, developing countries have gone through a profound economic transformation as they embark on a catch-up process with the advanced economies (Bourguignon, 2017). However, this process of convergence has been very unequal, with only a few countries (foremost China) undergoing a significant process of industrialisation (Rodrik, 2016), while vast parts of the developing world (notably South Asia and Africa) proceed at a much slower pace and follow a more fragile path (Lakner and Milanovic, 2015). As a result, the number of agricultural workers worldwide today is largely the same as it was 30 years ago (Figure 1). Despite the importance of land for the world’s poor people, we know almost nothing about its value and distribution, since the existing estimates reflect neither inequality of land ownership nor inequality of value and do not account for the landless population. As a result, both policy-makers and academic researchers lack basic information to evaluate the economic conditions and the ownership structure which dictate the lives of the world’s poorest people.

Precisely estimating land inequality is crucial given its relevance to debates that range from institutions and human capital accumulation to food security and poverty alleviation. The development literature has often emphasized the negative relationship between the initial unequal distribution of land and subsequent growth rates (Rodrik, 1995, Deininger and Squire, 1998, Kanbur, 2000, etc.) through various mechanisms. Unequal distribution of land adversely affects growth and development, as it results in institutions that preserve the distributive status quo (Engerman and Sokoloff, 1997; Sokoloff and Engerman, 2000). Land concentration hampers investment in education, as this goes against the interest of land elites (Galor, Moav, and Vollrath, 2009). It can also affect, and be reinforced by, poor financial development (binswanger1999explaining) as it restricts small actors’ access to credit, and their access to land markets. The resulting poor institutions, inadequate investment in education and provision of public goods, and underdevelopment of the financial market are some of the channels through which land inequality affects growth and development (Deininger and Squire, 1998; Easterly, 2007; Guereña and Wegerif, 2019).



Note: The upper left panel shows trends in the share of agricultural value added of total value added. The upper right panel depicts the share of agricultural employment of total employment. The bottom left panel corresponds to employment in agriculture in absolute numbers, and the bottom right panel shows trends in the share of the rural population of total population. These estimates are regional-weighted averages by country population sizes. Source: ILO and FAOStat.

Fig. 1. Global trends in agriculture since the 1990s

In this article, we introduce a new measurement approach of land inequality which accounts for the ownership aspect, the value of the land, and the landless population, exploiting household surveys. Following this methodology, we provide consistent estimates of inequality in land ownership across countries, and regions of the world, both in terms of area and value, as well as accounting for the landless population. This innovates and contributes to the existing literature in three ways. First, we focus on ownership rather than access to land or landholdings. Existing cross-country estimates of land inequality, and the ensuing literature which analyses its effect on economic outcomes, are based on the distribution of the size of operational holdings, constructed using agricultural census data (Taylor and Jodice, 1983¹;

¹For 54 countries in the 1960s, based on FAO World Census Agriculture

Deininger and Squire, 1998²; Frankema, 2010³). Operational holdings are defined as economic units of agricultural production under single *management* (FAO, 2018). This might be an adequate approach for efficiency concerns but pose serious conceptual challenges for measuring land inequality from an equity point of view since they do not capture the *ownership* of land holdings. Departing from the use of agricultural censuses, we exploit household survey data, which allows us to focus on land privately owned by a household rather than solely its holdings.

Second, this paper contributes to the literature on the value and productivity of land. Given that agricultural censuses do not capture the characteristics of the land, existing estimates do not account for differences in the *value* of land (e.g. due to soil quality or location). In the existing literature, the access to land has often been seen as an important policy tool for relieving poverty, underpinning the demands for the land redistribution reforms (Banerjee, 1999; Besley and Burgess, 2000; Griffin, Khan, and Ickowitz, 2002; Lipton, 2009). At the same time, sustained growth of agricultural productivity is a critical source of economic development (and poverty alleviation in the long-run; e.g. Eswaran and Kotwal, 2006, Gollin, Parente, and Rogerson, 2007, Restuccia, D. Yang, and Zhu, 2008, D. Yang and Zhu, 2013 etc), bringing up concerns about the effects of land redistribution on production efficiency and agricultural output. For example, the nature of the relationship between farm size and productivity is at the heart of the debate on desirability of land redistribution from the efficiency perspective (Sen, 1966; Berry and Cline, 1979; vollrath2007; Foster and Rosenzweig, 2017, etc.). Hence, there is a need to account not only for land sizes, but also for land values in inequality estimates. To the best of our knowledge, this is the first paper to present and explore the relationship between inequality in land area and inequality in land value.

Finally, since agricultural censuses do not capture landless households, this part of the population has been substantially disregarded in the literature and at best only roughly proxied. This is a significant shortcoming of the existing literature as variations in ownership rates across countries are unaccounted for. This section of the population are especially important since landless households are precisely the most vulnerable as they are at the bottom of the distribution but still rely on working and using land.

For all these reasons, it is unclear whether the distribution of the area of holdings from agricultural census data reliably captures overall land inequality. There is a need to assess the validity of this link and to better delineate the concept of land inequality that is most

²261 observations for 103 countries, based on FAO World Census Agriculture

³186 observations for 105 countries, based on census data from the International Institute of Agriculture (IIA) and FAO

pertinent in the context of developing countries. This paper is the first to provide comparable estimates of land inequality, under different definitions, in various countries across the world. We first examine the link between the area of inequality in holdings from censuses and our estimate of inequality in land ownership area from household surveys. We then provide estimates of both land area and value inequality among the landowning class.

Our results show that land value inequality differs significantly from land area inequality, and confirms the need to take both into account. Our results further show that accounting for the landless population increases land inequality unequally between countries due to differences in ownership rates. In addition, we provide a novel perspective on international patterns of land inequality. Our benchmark metric of agricultural land inequality (i.e. inequality of land value when including the landless population) reveals regional patterns that show South Asia and Latin America to be the most unequal regions of the world, followed by relatively more egalitarian African countries, and finally “Communist” Asia (China and Vietnam) as the least unequal world region. This paper thus makes an important contribution in showing the need for more critical use of existing estimates based on census data.

The rest of the article is organized as follows: In the next section, we discuss different concepts related to land inequality. We then describe the data and methodology used in this paper in section 3, followed by the main results in section 4.

2. Land inequality measurement: agricultural censuses vs. household surveys

The literature on land distribution has long relied on estimates of Gini coefficients for land using agricultural censuses, which provide tabulated data on the number of holdings and the total area of holding by size class (Deininger and Squire, 1998; Frankema, 2010). However, in this paper, we turn to household surveys instead. In what follows, we discuss what the measurement of land distribution captures when using the two data sources and why we prioritize surveys in this study.

First, land distribution calculated using an agricultural census captures the distribution of operational holdings, which are “an economic unit of agricultural production under single management comprising all livestock kept and all land used wholly or partly for agricultural

production purposes, without regard to title, legal form or size” (FAO, 2018). Hence, censuses do not directly capture the distribution of land ownership, since landholders may or may not be owners of the operational holding. Moreover, the census does not necessarily account for multiple landholdings per owner⁴, thus failing to capture the full extent of land concentration. On the other hand, household surveys often include an agricultural module which collects detailed information on land at the household level. The advantages of this source of data are numerous. Surveys provide a better idea of inequality in land ownership since each plot of land is linked to the household owning it, unlike with census data. Surveys also allow us to distinguish between privately-owned land and operated land i.e. land that is merely utilised by the household, for instance through renting or sharecropping.

The sparse literature relying on surveys has focused on the distribution of operated land. While this gives an idea of the extent of access to land in terms of utilisation, it is not equivalent to land ownership. In fact, households which operate land that they do not own will need to compensate the landowner for its use through rent payments or sharecropping. Moreover, land that is owned can also be used as collateral to gain access to credit or can be rented out or sold in case of a need for liquidity – hence the need to distinguish between land that is merely operated and land that is effectively owned (Doss et al., 2015).

Second, census data typically cover not only household farms but also government lands, and holdings by private corporations.⁵ Communal land is not considered an operational holding and is, therefore, not captured by census statistics. The ample coverage of farms, regardless of the landholder being a household or not, is an advantage of the census to analyse the productive structure of the agricultural sector. It is less clear how to interpret statistics of land inequality which mix household holdings with public and corporate ones (note, also, that corporations can be owned by numerous households and by both the government and foreigners).

Household surveys, by contrast, do only capture land owned by households. Like cen-

⁴“The holding’s land may consist of one or more parcels, located in one or more separate areas or in one or more territorial or administrative divisions, providing that they all share such means of production as labour, farm buildings, machinery or draught animals. Several different economic agricultural production units under the same ownership, or under the same general management, may be considered as separate holdings if they are operated by different persons.” (FAO, 1999). As explained by vollrath2007, the distribution of operational holdings does not capture the distribution of land ownership. The distribution of land holdings is relevant if “we are interested in efficiency, not equity”.

⁵The sector coverage, however, varies across countries and over time. For example, most African countries only cover land operated by the household sector (excluding corporate land).

suses, surveys often do not include communal land. One caveat of the survey is that it misses land that is not owned directly by households but indirectly through businesses, potentially underestimating the actual level of inequality (since equity ownership generally concentrates at the top of the distribution). [Lowder, Scoet, and Raney, 2016](#) estimate that family farms *operate* around 75% of total agricultural land, indicating that households are responsible for most of the world’s agricultural and food production. To the extent that land *operated* and land *owned* are tightly connected, this would suggest a large role for households in agricultural land. In the results section we show that for the sub-sample of countries included in our study for which this information is available, this figure is above 80 percent in most cases.

Third, differences in the value and quality of land are not measured in agricultural censuses. Unlike agricultural censuses, household surveys often provide information on the area (e.g., GPS measurements and farmers’ estimates) as well as the market value of land at the household level. Both are valuable information as the distribution of land in terms of area might not be equivalent to the distribution in terms of value. For instance, if larger landowners have disproportionately more valuable land, then land area inequality would not reflect the full extent of the unequal distribution. This paper bridges the gap in the literature when it comes to land value inequality and provides consistent estimates across countries.

Additionally, census data, by definition, do not account for landless households. This omission is crucial as the share of the landless population vary in different countries and thus existing estimates may not be fully comparable across countries. For instance, based on estimates of inequality between landowning households, a country where land is equally distributed among only a handful of landowners will have a lower level of inequality compared to another country that has a more disparate distribution of land ownership among a larger share of landowning households. There is thus a need to include landless households to account for the full picture. In fact, [Erickson and Vollrath, 2004](#) show that the established effect of land inequality on institutions and financial development is sensitive to the inclusion of the landless population.

[Erickson and Vollrath, 2004](#) propose a complementary measure of inequality, which is the ratio between agricultural population and the number of holdings; this aims to capture the extent to which holdings are spread across the relevant population, using FAO data. However, the implicit assumption behind such a proxy for landless households is that each agricultural holding has a single owner. Although it is an improvement vis-à-vis the existing literature, this raises concerns similar to those around the existing literature on land

inequality.

Finally, as argued by [Lowder, Scoet, and Raney, 2016](#), the coverage and methodology of agricultural censuses are not uniform between countries and over time, especially in developing countries, despite efforts by FAO to encourage uniformity. Agricultural censuses in different countries do not systematically distinguish between different forms of legal ownership and can also have different minimum thresholds for recording holdings, which further reduces comparability. Household surveys, on the other hand, provide the flexibility required to make them more comparable across countries and over time. Some papers in the literature have turned to household surveys to assess land distribution in different countries (see [Doss et al., 2015](#) for a review on gendered land outcome in Africa based on surveys).

These factors suggest that agricultural census data do not allow to grasp the full extent of land inequality. Despite the caveats of survey data, surveys remain a substantially better source when estimating inequality of land ownership. Surveys are mostly nationally representative and hence effectively designed to capture all types of households, whether landowning or landless. They provide detailed data on the land owned by a household, which allows for an in-depth analysis of inequality in land ownership in terms of both area and value, while accounting for the landless population. To the best of our knowledge, this paper represents the first attempt to provide a comprehensive estimation of the distribution of landownership inequality by both area and value that is comparable across countries, spanning different continents, by exploiting household surveys.

3. Data & Methodology

This section starts revisiting and updating estimates of land area inequality based on agricultural census data. This data source is centralised and overseen by the UN Food and Agriculture Organization (FAO) and is published at the country level every decade under the World Programme for the Census of Agriculture (WCA). Agricultural census reports provide a tabulated distribution of operational holdings by size bracket.⁶ Previous estimates of land distribution based on this source cover most of the twentieth century, but with only a few estimates in the early 2000s ([Deininger and Squire, 1998](#); [Frankema, 2010](#)). In this paper, we reassess and update estimates of land inequality based on census data, up to the most recently available year (which generally correspond with the 2010 decennial round).

⁶Some countries further provide decompositions by tenure, gender, land use and crops.

Given the tabulated format of the data, we use the generalised Pareto interpolation method (Blanchet, Fournier, and Piketty, 2017) to update census-based estimates of inequality. The goal is to examine census-based estimates relative to survey-based ones.

As explained previously, we then use household surveys to provide estimates of land area and value distribution, as well as including the landless population in different countries across the world. Two main types of surveys are used in this paper to do so: the World Bank’s Living Standards Measurement Study (LSMS) household surveys and official household surveys conducted by different countries.⁷ These generally comprise of an agricultural module which collects information about the fields or plots owned by the household. The relevant information for estimating land ownership inequality is the land area, reported value and an indication of ownership.

The choice of countries in this paper is based on the availability of household surveys capturing the ownership of land (Table A1). In some countries the quality of the data was not sufficiently good and they were therefore excluded from the analysis. Most surveys available in the different countries have a very short temporal dimension (e.g. in several cases only one year of data is available). For this reason, we restrict our analysis to a single observation per country and do not analyse trends in the concentration of agricultural land.

Our object of analysis is to measure the distribution of land ownership. In this paper, land ownership is defined as any agricultural land over which a household has private property rights. This is defined fairly consistently across countries. China and Vietnam are special cases, as in these countries private property is less clearly defined but rural households are given extensive rights over land e.g. rights to control, profit from, and inherit land (McKinley and Griffin, 1993; Li and Zhao, 2007; Do and Iyer, 2003; Piketty, L. Yang, and Zucman, 2019). For further discussion on this, please refer to Appendix D.

As in previous studies that rely on census data, household surveys unfortunately also do not account for access to communal land. Communal land is often large plots of land that can be used equally by the members of the community to which it belongs. Members typically do not have exclusive or formal rights on the plot of land: they normally have

⁷In addition, we use the Demographic and Health Surveys (DHS) of the DHS Program for robustness purposes. DHS provides two key pieces of information regarding land, a direct question on ownership and the area of land owned, but there are no estimates of the value of the land. We thus estimate land ownership inequality and obtain estimates close to our main estimates, using the LSMS and other household surveys (see Figure C1).

use rights and sometimes common control rights, and very rarely transfer rights. Beyond the conceptual challenges of land ownership when it comes to communal land, developing countries, especially in Africa, also face a more generalized issue of lack of land registration. This poses challenges in terms of securing land rights, and access to credit. [Deininger and Feder, 2009](#) document the impact of land administration interventions and conclude that land registration is beneficial to ensure a secure land tenure system and for the development of a formal land market, which is associated with higher investment, productivity, and overall efficiency. In addition, the formalization of these land provides better data that is crucially needed to analyse the importance of communal land in different countries.⁸

In this paper we focus on two ways of measuring the agricultural land owned by a household. The first is in terms of area of agricultural land (i.e. the size of the land holdings owned by a household).⁹ The second is in terms of the value of agricultural land. The latter is our preferred measure since it accounts for the large heterogeneity of land types within a country and captures the value of land as an asset. Values reported by surveys are based on the concept of current market value, where agricultural land is valued at prevailing market prices.¹⁰

To describe the distribution of agricultural land, we use standard measures of inequality such as the Gini coefficient and land shares (i.e. the percentage of land owned by a population group such as the top 10%, middle 40%, or bottom 50%). Although the Gini coefficient has been used predominantly in land inequality studies based on census data, we prefer to use land shares. The Gini index is a synthetic measure of inequality which summarises the entire distribution into a single number, and it is thus less informative about where the important changes in distribution take place.

We measure inequality of land ownership within two population groups. The first group

⁸The success case of Mexico in its registration of communally-owned land provides a case study of the possibility of formalizing these lands and the potential gains in doing so. It also entails the availability of better data to study communal land ([Morett-Sánchez and Cosío-Ruiz, 2017](#)).

⁹Note that agricultural land area is reported in both agricultural censuses and in surveys. The difference is that surveys measure land ownership at the household level while agricultural censuses measure the land area of operational holdings.

¹⁰The valuation practice in surveys is generally based on a subjective assessment of respondents (surveys generally ask a question along the lines of: “What would be the amount received if the land was sold today?”), but this is often complemented by external assessments based on administrative data. In certain instances, in particular in the absence of well functioning agricultural land markets, the survey design evaluates the market value of land using alternative approaches, such as by capitalising agricultural income (for example, this is an approach adopted by the China Family Panel Studies (CFPS)).

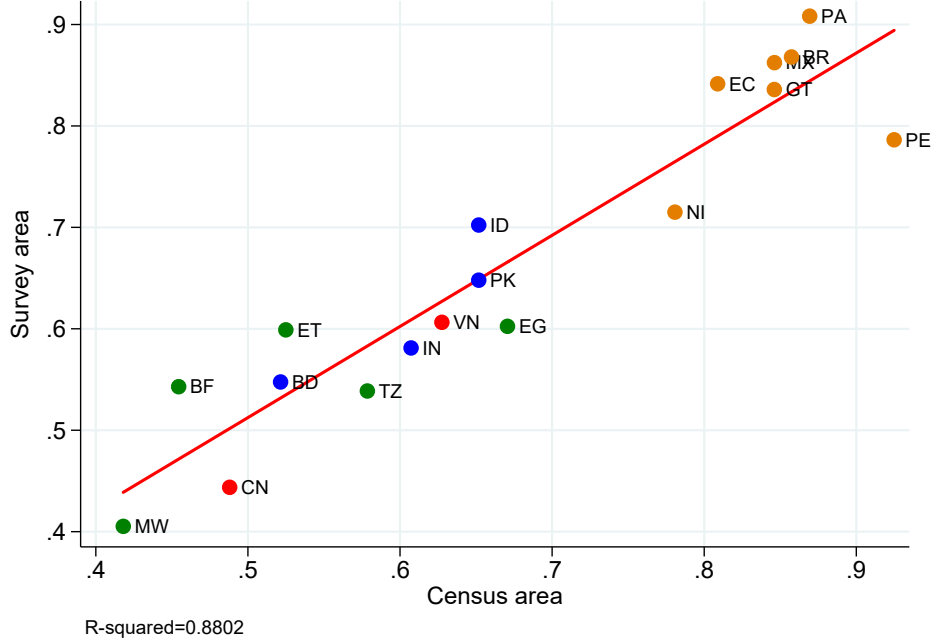
consists of landowners (i.e. households owning land). Our second group consists of landowners plus landless households. The latter is our benchmark unit, since it is important to account for landless households to obtain a complete picture of land inequality. Surveys are extremely useful in examining the landless population, since they capture both the population of households living in rural areas and the working activities of each member of a household, including agriculture. This information, together with the number of households who are landowners, allows us to identify the population of ‘landless households’. We define landless households as those where at least one of its members is employed in agriculture but does not report owning any agricultural land.

4. Results

4.1. *Land area inequality: census vs. survey*

Previous work on the measurement of the distribution of land ownership across countries has been based on information contained in agricultural censuses. However, as argued above, censuses do not record agricultural ownership units but rather operational (or production) units. The implicit assumption behind this is that the size distribution of operational holdings provided in censuses serves as a proxy for the distribution of land ownership. Moreover, the use of census data has restricted analysis to inequality of land area and not of land value, as well as to inequality between landowners, excluding landless households.

Given the wide use of census-based estimates in the literature, as a first step it is useful to examine the extent to which the size distribution of farms reflects the distribution of land area ownership. Figure 2 compares agricultural land inequality estimated from survey and census data. More precisely, it shows the Gini index for the distribution of land area: (i) among households owning land from surveys (x axis), and (ii) among land holdings from agricultural censuses (y axis). In order to ensure comparability, we select rounds of survey data that are the closest to the census year of the countries included.



Note: This figure includes Mexico, Peru, Egypt, and Burkina Faso, for which we have land area estimates from surveys but no information on value. They are hence not part of the sections that follow. Conversely, the Gambia, Nigeria, and Niger do not appear in this figure as there is no census information on the distribution of holdings. In order to ensure comparability, we select rounds of survey data that are the closest to the census year of the respective country.

The countries and the year of survey are as follows: BD – Bangladesh (2011); BF – Burkina Faso (2014); BR – Brazil (1996); CN – China (2002); EC – Ecuador (2014); EG – Egypt (2018); ET – Ethiopia (2011); GT – Guatemala (2000); ID – Indonesia (2014); IN – India (2012); MW – Malawi (2010); MX – Mexico (2009); NI – Nicaragua (2014); PA – Panama (2008); PE – Peru (2007); PK – Pakistan (2010); TZ – Tanzania (2018); VN – Vietnam (2014). For sources of data, see Appendix A.

Fig. 2. Gini index based on census and survey data in selected countries

Interestingly, the Gini index is broadly comparable according to the different definitions in the two data sources (the regression line is almost equivalent to the 45-degree line). We find, according to both sources, that land inequality is highest in Latin America, is at an intermediate level in Asia, and is lowest in Africa.

Given that the two estimates of land inequality tend to coincide, this suggests that inequality of landholding area can be an appropriate proxy for inequality of land area ownership. The high correlation between the two sources should, therefore, reflect the following three points: (i) households own the vast majority of the existing agricultural land; (ii) most of the landholders are also landowners; (iii) landowners do not own a significant amount of

land through multiple operational holdings.

Making use of the information available in the latest two rounds of FAO censuses (2000 and 2010), we can provide evidence supporting the first two points (unfortunately, we could not find data on the third point). First, we can observe the total amount of land in operational holdings that is owned by landholders. This information is available for a large number of countries, both in rich and developing regions (table appendix E1). Interestingly, we find that on average, over 80% of the operated land in Asia, Latin America and Africa is also owned by the landholder (this is in contrast with Europe, North America and Oceania, where this figure is between 50 and 65%). Second, in a smaller set of countries, censuses also report the legal status of the landholder, distinguishing between civil and juridical landholders (the first status largely captures households¹¹). Within the sample of countries included in this study, in eight cases this information was available (figure appendix E1). In seven countries, the amount of land operated by households is over 80 percent. The only exception is Peru, where the share is closer to 40 percent. This could potentially explain why results for Peru are slightly more distant from the regression line than other countries in Figure 2.

However, despite the similar Gini estimates, the various caveats associated with census data, such as inconsistencies in terms of coverage (household, corporate or government sector included or not, in an unsystematic way) should be kept in mind. Additionally, while census data could be seen as a first approximation for inequality in land area, it does not reflect inequality in land value. Section 4.2 expands on this by including different dimensions of inequality to arrive at our benchmark concept of inequality, which is that of the distribution of agricultural land value among agricultural households (including both landowners and landless households).

4.2. *Distribution of land area vs land values*

The value of one hectare of agricultural land can vary widely within a country, with numerous factors explaining the differences: diversity in soil quality (Benjamin, 1995), type of agricultural cultivation (e.g. cropland vs pastures), access to irrigation and agricultural capital, area of the agricultural holding (Barrett, 1996; Martinelli, 2016), land markets reg-

¹¹The classification of the legal status of the landholder varies across countries. The most common categories are civil vs. juridical, but some countries use different legal classifications. For example, in Brazil, the legal status is decomposed into seven categories: (i) Individual, (ii) Condominium, consortium or partnership, (iii) Cooperative, (iv) Limited liability company, (v) Institution, (vi) Government, and (vii) Others.

ulation (Restuccia and Santaaulalia-Llopis, 2017), factor market imperfections (Sen, 1966), etc. It is clear that agricultural land is not a homogeneous asset and that estimates of land-area inequality fail to capture the diversity of values across land holdings.

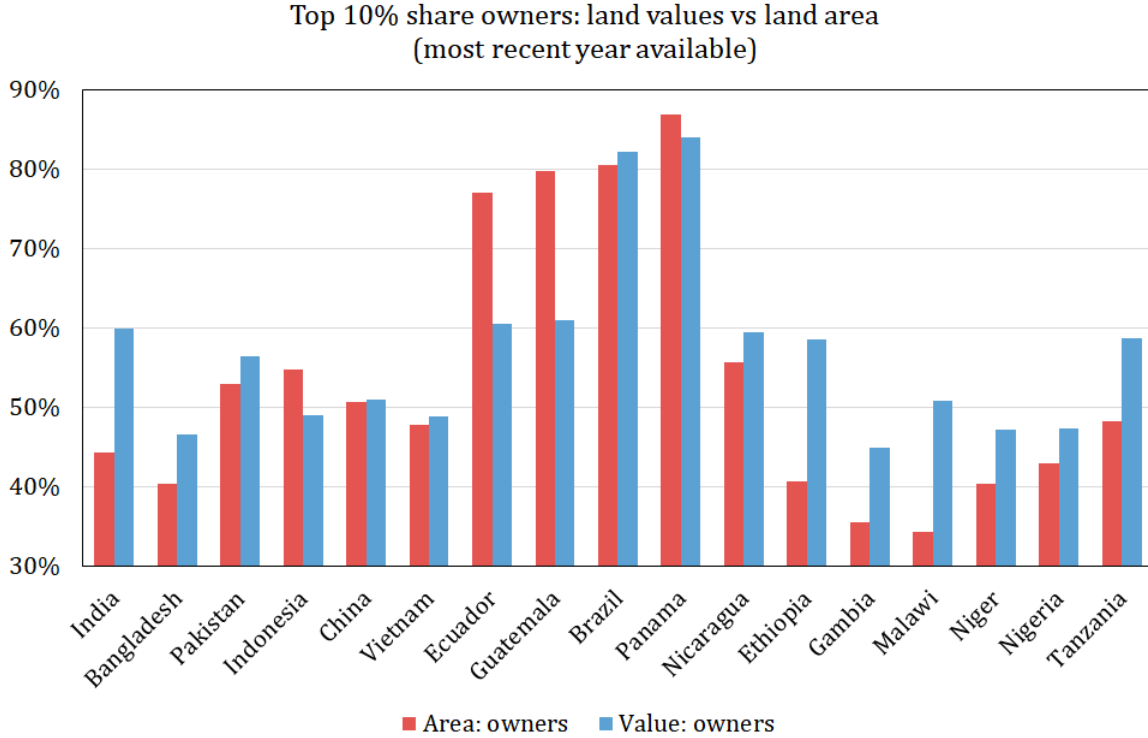
We go one step further than previous studies and compare inequality of land area with that of land values in household surveys, whenever this information is available. Figure 3 shows the distribution of agricultural land within landowning groups for the most recent year for which data are available, using two measures of agricultural land: land area (red bars) and land values (blue bars). In other words, the figure shows the share of total agricultural land owned by the top 10% of landowners, according to the two measures.

Importantly, the results indicate that inequality of land value can be significantly different from inequality of land area. For example, a comparison of India and Ethiopia with Ecuador and Guatemala is particularly informative. The first two countries have relatively low levels of land area concentration when compared with the second two: in fact, the share of the top 10% of landowners in Ecuador and Guatemala is twice that of the same groups in India and Ethiopia.¹² From this perspective, inequality in these two pairs of countries is remarkably different. Based solely on land area estimates, Ethiopia and India would be assessed as being relatively egalitarian countries compared with Ecuador and Guatemala, which are extremely unequal by any standard. However, results for land value inequality, as opposed to land area inequality, completely change this comparison. Under the land value metric, differences between the four countries all but disappear, as the top 10% of landowners own around 60% of total agricultural land value in all four countries.

Generally, our results point to important differences between land value inequality and land area inequality. In Appendix Figure E2, we repeat the comparison between the Gini index in the census and the survey, but this time between land area in census and land value in surveys. While we still find a positive correlation between the two measures, it is much lower than in Figure 2. Concretely, the R-squared drops from 0.88 to 0.55, confirming that the distribution of land area from censuses only midly predicts the distribution of land values.¹³

¹²More precisely, 80% of total agricultural land is owned by the top 10% of landowners in Ecuador and Guatemala compared with 40% in India and Ethiopia.

¹³The sample of countries in Figure 2 and appendix figure E2 is not exactly the same. In appendix figure E3 we replicate figure 2 using only the countries that are also included in figure appendix E2. In this case, the drop in the R-squared is even larger: from 0.92 to 0.55.



Note: This graph provides estimates of the top 10% share of area and value among the owning class, both from the urban and rural area.

Fig. 3. Share of top 10% of land owners in selected countries, by land area and land value

4.3. Accounting for the landless population

As explained in previous sections, a meaningful measurement of the distribution of agricultural land should not be restricted solely to landowners. While inequality within the landowning class provides useful insights into the structure of inequality, a comprehensive assessment of the phenomenon needs to include the landless population (i.e. those directly involved in agriculture but who do not own land).

Figures 4a and 4b show the land shares of the top 10% and bottom 50%, respectively, for land value among landowners (blue bars) and landowners and landless households (green bars). In addition, Table 1 shows the share of landless households within the population of landowners plus landless households. Figures 4a and 4b show that including landless households is important in establishing levels of inequality. More specifically, regions with the highest shares of landless households show larger increases in levels of inequality.

Three patterns are worth mentioning: first, South Asia (India, Bangladesh, Pakistan and

Indonesia) and Latin America (Nicargagua, Ecuador, Guatemala, Brazil and Panama) are the most unequal regions, with the share of the top 10% rising to up to 70-80%, and that of the bottom 50% falling from 7–10% to 0–2%. In both regions, landless households account for more than one-third of the reference population.

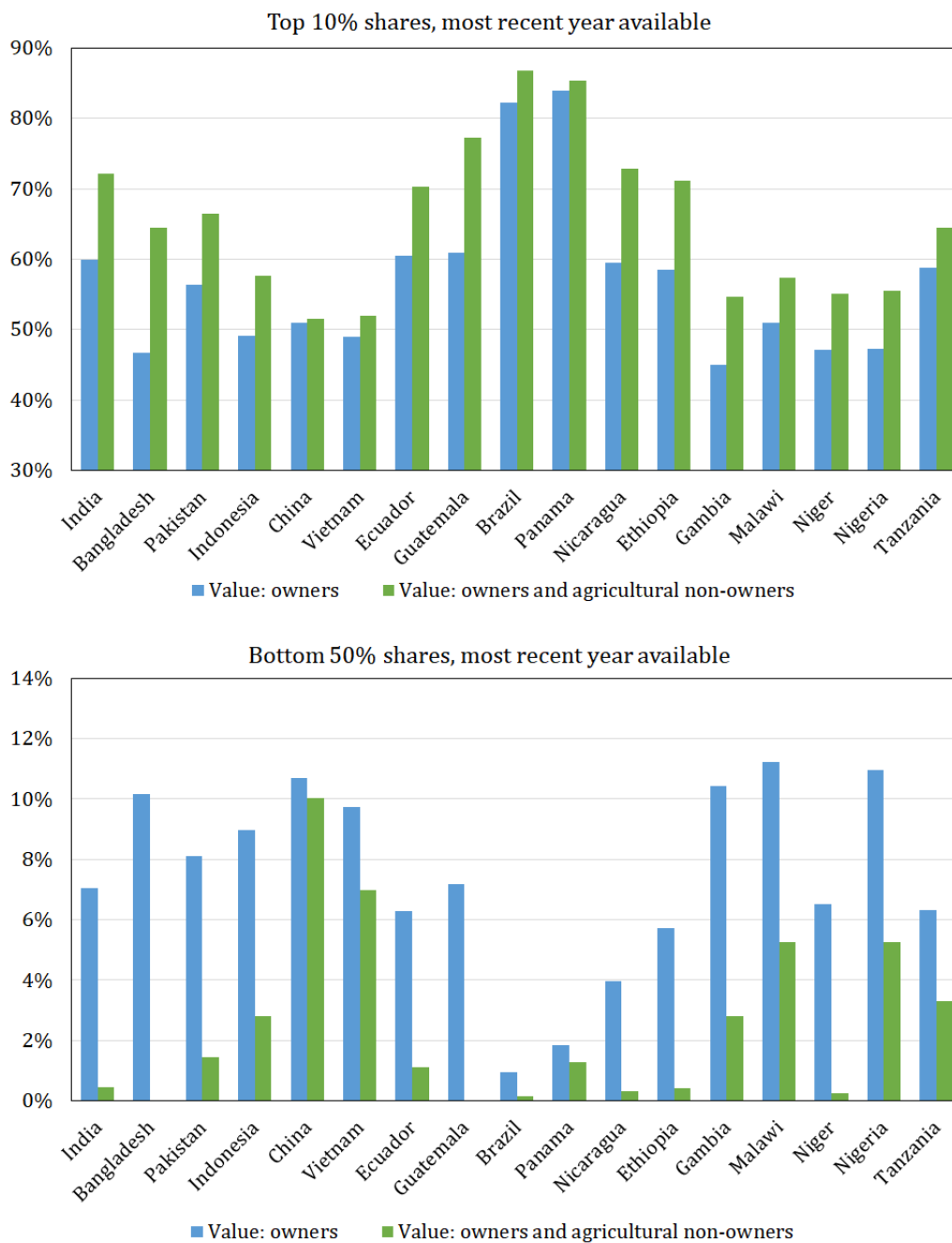


Fig. 4. Fig. 4a (upper panel) and Fig. 4b (lower panel) : Shares of land value in selected countries of top 10% and bottom 50% of land owners, and land owners plus the landless

In contrast, inequality in China and Vietnam is not significantly affected by the inclusion of the landless population, with an increase of a few percentage points in the share of the top 10% (and a decrease in the share of the bottom 50%). This is driven by the very low proportion of landless households, around 3–12%. This is explained by historical land reforms carried out in these countries under their communist regimes, which still provide widespread access today to agricultural land for most households in rural areas.

Finally, African countries have proportions of landless households that are somewhere in between. Hence, changes in levels of inequality when switching from one population concept to the other are in the middle compared with the two groups of countries as well.

Landless households as a percentage of agricultural households						
World Regions	Individual Countries					
South Asia 36%	India 39%	Bangladesh 40%	Pakistan 36%	Indonesia 28%		
China-Vietnam 7%	China 3%	Vietnam 12%				
Sub-Saharan Africa 29%	Ethiopia 40%	Gambia 33%	Malawi 25%	Niger 27%	Nigeria 27%	Tanzania 21%
Latin America 35%	Ecuador 36%	Guatemala 56%	Brazil 29%	Nicaragua 40%	Panama 12%	

Table 1: Proportion of landless households

Note: This table provides the proportion of landless household out of the landowning and landless households. The household is defined as landless i) if it does not own any piece of land and ii) if at least one household member participates in the agriculturally-related activities. We include Brazil and Peru in this table since we observe the percentage of landless households, despite surveys not covering the value of land. Hence, they are not included in subsequent analyses.

Overall, it is clear that any assessment of land inequality that excludes the landless population will result in an incomplete understanding of the complex structure of inequality present in different countries.

4.4. *From land area inequality between landowners to accounting for landless people and land values*

Figures 5a and 5b summarise the main results of the paper. They show the agricultural land shares of the top 10% and the bottom 50% for the three concepts examined in this paper: (i) land area inequality between landowners; (ii) land value inequality between landowners; and (iii) land value inequality within the population of landowners plus landless households. Rather than presenting results at the country level (as done in previous sections), Figures 5a and 5b show the unweighted country averages for four world regions: (i) South Asia – Bangladesh, India, Pakistan, Indonesia; (ii) China and Vietnam; (iii) Latin America – Nicaragua, Ecuador, Guatemala, Brazil, Panama; (iv) Africa: Ethiopia, the Gambia, Malawi, Niger, Nigeria, and Tanzania. The country groupings are based not only on geographical location but also on common patterns in the ownership of agricultural land and in macroeconomic trends (e.g. proportion of employment and value added in agriculture; share of rural population, etc.).

Figures 5a and 5b condense the main patterns documented in the paper. First, countries in the South Asia region appear to be moderately equal when looking at the distribution of land area between landowners. However, these countries have some of the highest levels of inequality when land values and the landless population are included. China and Vietnam, by contrast, display higher levels of land area inequality between landowners than both South Asia and Africa, but land concentration is only slightly higher when land values and landless households are taken into account. Overall, China and Vietnam appear to be the least unequal world region in our sample, according to our benchmark indicator of inequality.

Latin America (at least as reflected by Nicaragua, Ecuador, Guatemala, Brazil, and Panama) displays the most unequal distribution of agricultural land area between landowners. This also applies to Mexico and Peru (Figure 2), and is a finding that has been documented in most Latin American countries based on agricultural censuses (Frankema, 2010). Unlike the other world regions, inequality between landowners is somewhat lower in land value than in land area, a trend driven by three of the five countries: Ecuador, Guatemala, and Panama. When the landless population is included, similar patterns of land inequality are observed, with land value inequality also at one of the highest levels.

Finally, the African countries selected for this analysis occupy an intermediate position between China and Vietnam on the one hand and South Asia and Latin America on the

other. Africa has the lowest levels of land area inequality between landowners, but inequality rises gradually when land values and the landless population are included.

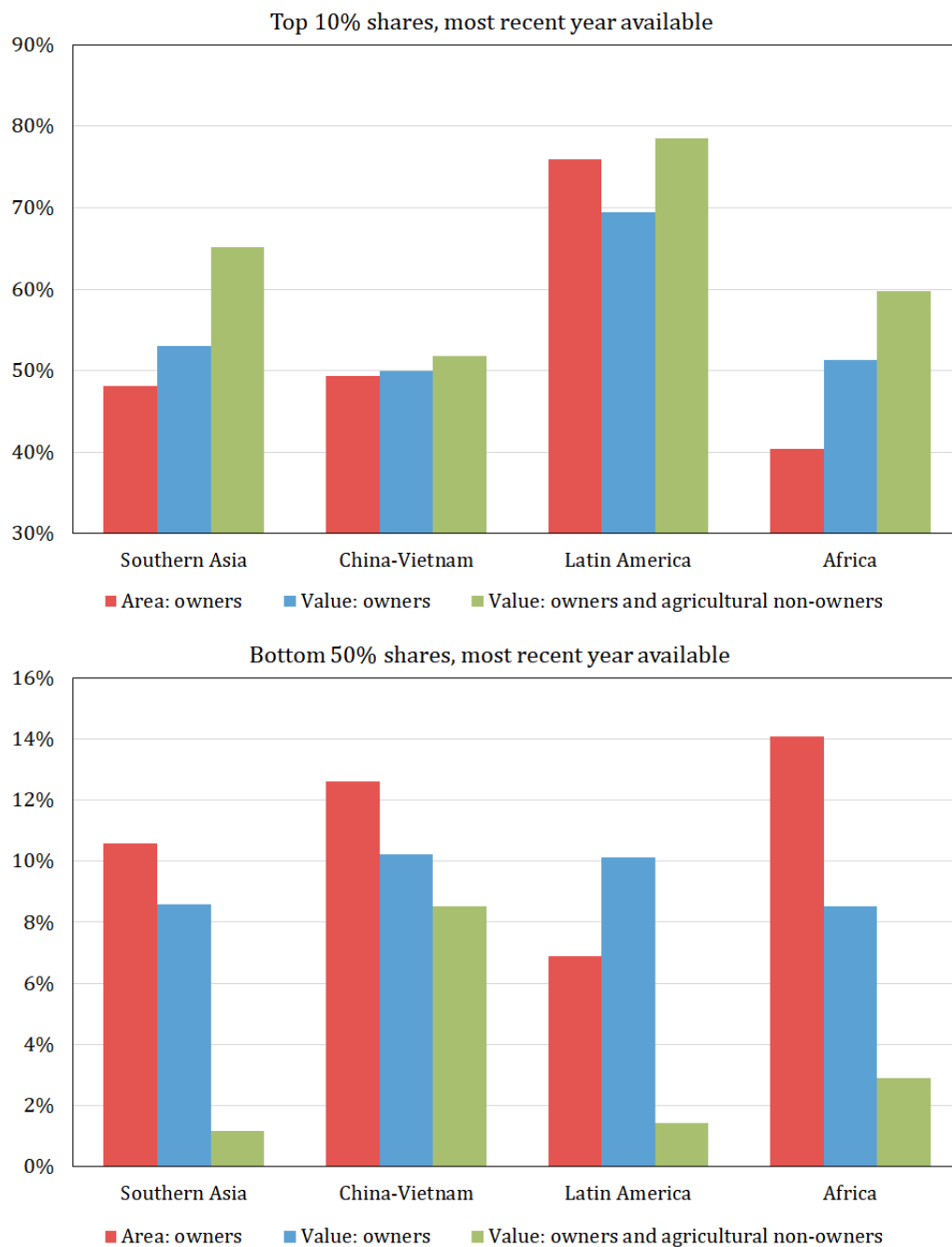


Fig. 5. Fig. 5a (upper panel) and Fig. 5b (lower panel): Shares of land area and land value held by top 10% and bottom 50% of landowning class and including landless households, in selected countries

4.5. *Conclusion and Next Steps*

This paper provides the first consistent estimates of agricultural land inequality in developing countries. As such, it presents the most comprehensive overview of the different dimensions of inequality in agricultural land and emphasises the importance of using well defined concepts and clear methodology for measurement. Notably, it shows that we need to go beyond existing studies looking at the size distribution of agricultural holdings based on agricultural censuses. Existing estimates do not reflect either land ownership inequality or land value inequality and do not account for the landless population. We advocate instead the use of household surveys as the most appropriate data source to estimate land ownership inequality across countries, both in terms of area and value, and to account for non-owners.

Our new estimates provide a novel perspective on international patterns of agricultural land inequality. According to our benchmark metric (i.e. land value inequality including the landless population), South Asia and Latin America show the highest levels of inequality, with the top 10% of landowners capturing up to 75% of agricultural land and the bottom 50% owning less than 2%. The African countries selected display relatively less unequal land ownership patterns, while “Communist” Asia (China and Vietnam) represent the region with the lowest levels of inequality.

Having said this, we need to stress that the current results represent a first attempt at assessing agricultural land distribution in developing countries. Although we have included the most populated countries in the analysis, we intend to cover more developing countries to obtain a more complete picture. To do this, we are also developing robust approaches to estimate land values in countries for which surveys provide information on land area only (e.g. Mexico, Egypt, Russia).

To finish, we indicate several methodological extensions of the current work. First, we need to critically assess the role of different forms of land ownership, especially those for which distinctions from private property are not clear-cut (e.g. such as the role of communal land). Related to this, we need to better understand the importance of corporate land and public land and its impact on distributional patterns. In the future, an effort will be made to combine survey data and census data.

Finally, given the importance of land for the world’s poorest people, we stress the need for governments and international organisations to invest more in collecting more detailed and systematic information on agricultural land in household surveys, especially in countries

where data are currently not available.

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Appendix A. Sources of Data

Data sources census and surveys				
Country	Type	Year	Description	Source
AFRICA				
Burkina Faso	Census Survey	2010	FAO 2010	FAO
	Census Survey	2014	Enquete Multisectorielle Continue (EMC-BF) 2014	LSMS (World Bank)
Egypt	Census Survey	2009	FAO World Census of Agriculture 2010	FAO
	Census Survey	2018	Egypt Labor Market Panel Survey	Egypt's Central Agency for Public Mobilization and Statistics
Ethiopia	Census Survey	2000	FAO World Census of Agriculture 2000	FAO
	Census Survey	2015	Ethiopia Socioeconomic Survey (ESS)- Wave 3 (2015-16)	LSMS (World Bank)
Gambia	Census Survey	2015	Gambia Integrated Household Survey (IHS) 2015	The Gambia Bureau of Statistics
Malawi	Census Survey	2006	FAO World Census of Agriculture 2010	FAO
Niger	Census Survey	2016	Fourth Integrated Household Survey (IHS4) 2016	LSMS (World Bank)
	Census Survey	2014	National Survey on Household Living Conditions and Agriculture 2014	LSMS (World Bank)
Nigeria	Census Survey	2015	General Household Survey, Panel 2015-2016	LSMS (World Bank)
Tanzania	Census Survey	2007	FAO World Census of Agriculture 2010	FAO
	Census Survey	2014	National Panel Survey (NPS) 2014-15	LSMS (World Bank)
ASIA				
Bangladesh	Census Survey	2008	FAO World Census of Agriculture 2010	FAO
China	Census Survey	2011	Bangladesh Integrated Household Survey (BIHS) 2011	International Food Policy Research Institute (IFPRI)
	Census Survey	2002	ILC	Khan 2001
India	Census Survey	2002	Chinese Household Income Project	Chinese Academy of Sciences and others
	Census Survey	2016	China Family Panel Studies (CFPS) 2016	Institute of Social Science Survey, Peking University
Indonesia	Census Survey	2010	FAO World Census of Agriculture 2010	FAO
	Census Survey	2012	All India Debt and Investment Survey (AIDIS), 2012	Ministry of Statistics and Programme Implementation
Pakistan	Census Survey	2013	FAO World Census of Agriculture 2010	FAO
	Census Survey	2014	Indonesia Family Life Survey	FAO
Vietnam	Census Survey	2010	FAO World Census of Agriculture 2010	FAO
	Census Survey	2011	Pakistan Household Integrated Survey (HIES), 2010-11	Pakistan Bureau of Statistics
	Census Survey	2014	FAO World Census of Agriculture 2010	FAO
	Census Survey	2014	Vietnam Household Living Standards Survey (VHLSS), 2014	General Statistical Office (GSO) of Vietnam
LATIN AMERICA				
Brazil	Census Survey	1996	FAO World Census of Agriculture 2000	FAO
Ecuador	Census Survey	1997	Pesquisa sobre padrões de vida 1996-1997	LSMS (World Bank)
	Census Survey	2000	FAO World Census of Agriculture 2000	FAO
Guatemala	Census Survey	2014	Ecuador Living Conditions Survey	National Statistical Office of Ecuador
	Census Survey	2003	FAO World Census of Agriculture 2000	FAO
Mexico	Census Survey	2000	Encuesta Nacional sobre Condiciones de Vida 2000	LSMS (World Bank)
	Census Survey	2007	FAO World Census of Agriculture 2010	FAO
Nicaragua	Census Survey	2009	Mexican Family Life Survey	UIA and CIDE
	Census Survey	2011	FAO World Census of Agriculture 2010	FAO
Panama	Census Survey	2014	Encuesta Nacional Sobre Medicion de Nivel de Vida	National Statistical Office of Nicaragua
	Census Survey	2011	FAO World Census of Agriculture 2010	FAO
Peru	Census Survey	2008	Encuesta de Niveles de Vida	LSMS (World Bank)
	Census Survey	2012	FAO World Census of Agriculture 2010	FAO
	Census Survey	2007	Encuesta Nacional de Hogares sobre Condiciones de Vida y Pobreza 2007	National Statistical Office of Peru

Table A1: Sources of Data

Appendix B. Results

Country	Bottom 50%	Middle 40%	Top 10%	Gini index
India (2012)	0,5%	27,4%	72,1%	0,84
Bangladesh (2015)	0,0%	31,5%	68,5%	0,84
Pakistan (2010)	1,4%	32,2%	66,4%	0,80
Indonesia (2014)	2,8%	39,5%	57,7%	0,74
China (2012)	10,0%	38,4%	51,5%	0,64
Vietnam (2014)	7,0%	41,1%	51,9%	0,68
Ecuador (2014)	1,1%	28,6%	70,3%	0,82
Brazil (1997)	0,1%	13,1%	86,7%	0,91
Guatemala (2000)	0,0%	22,7%	77,3%	0,88
Nicaragua (2014)	0,3%	26,8%	72,9%	0,85
Panama (2008)	1,3%	13,4%	85,4%	0,90
Ethiopia (2015)	0,4%	28,4%	71,2%	0,83
Gambia (2015)	2,8%	42,5%	54,7%	0,73
Malawi (2016)	5,3%	37,3%	57,4%	0,72
Niger (2014)	0,3%	44,6%	55,2%	0,75
Nigeria (2015)	5,3%	39,1%	55,6%	0,71
Tanzania (2015)	3,3%	32,3%	64,4%	0,77

Note: Distribution of agricultural land value, including the landless population

Table B1: Agricultural land distribution

Appendix C. DHS and LSMS comparison

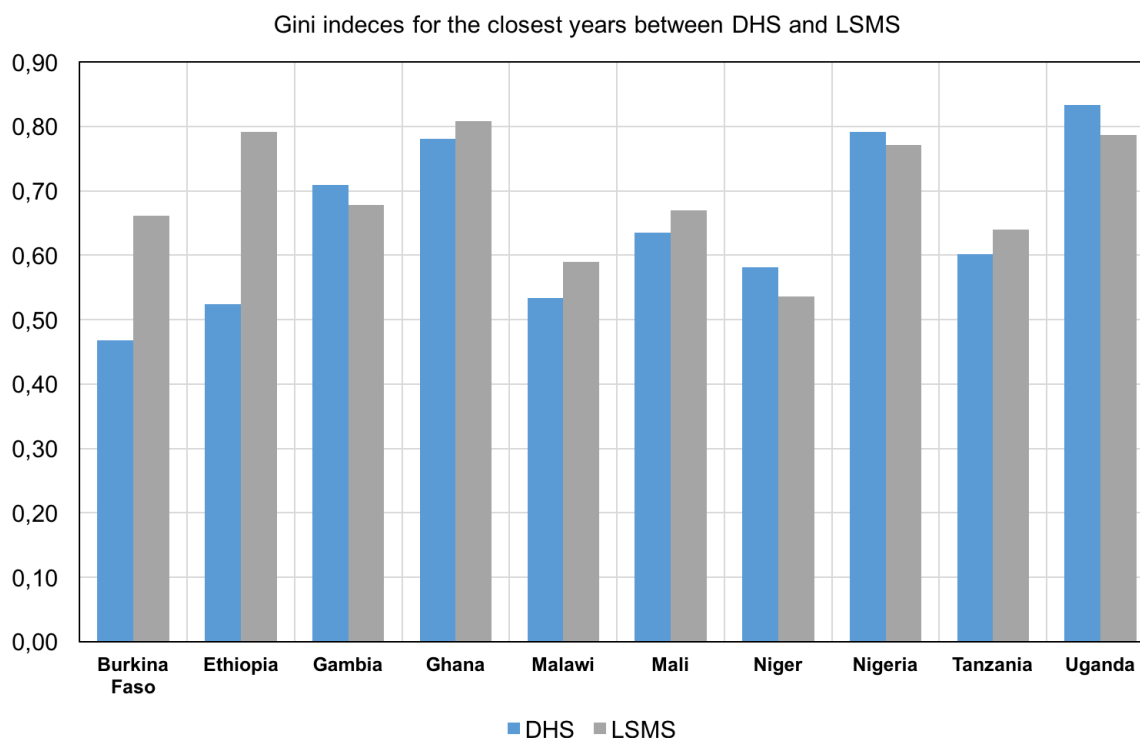


Fig. 6. **Figure C1:** Gini indices among owners based on DHS and LSMS

As part of the effort by the World Bank, the Living Standards Measurement Surveys (LSMS) has been implemented in a number of countries, aiming at providing nationally-representative household surveys and in some countries, with a panel component. The coverage of the LSMS is particularly wide in Africa compared to other world regions, providing detailed information on agricultural activities and land. Since the focus of these surveys often aim at capturing agricultural activities, they cover both land operated or owned by households. In surveys in which the distinction between the two are not straightforward, a proxy for ownership is defined as individuals who have inherited or purchased land. As a robustness check, the Demographic Health Surveys (DHS) are used. These are nationally-representative household surveys that focus on health and nutrition aspects but also have basic information on land ownership since the 2000s, reporting whether a given household owns or not land, and the area of the land owned. Gini coefficients estimated from the LSMS and DHS being very similar, validating the ownership proxy of the LSMS (Figure B1).

Appendix D. Ownership Regime

In order to construct commensurate land inequality estimates across various countries, and more importantly, regions of the world, differences in land ownership regimes must be taken into account. Ideally, all surveys would contain a question directly asking, “The household enjoys which rights over this plot of land?”, followed by an extensive list of all possible property rights (i.e. right to keep profits from land, right to transfer land, right to sell land or use as collateral, etc) which exist in each country. This would allow for the creation of subsamples of households which have exactly equivalent ownership regimes across countries which may have different institutional setups, resulting in inequality measures which are systematically comparable. In the absence of such detailed data, it is necessary to understand the exact land rights that households used in our analysis actually possess.

With this goal, we identify the key rights associated with the economic benefits of land ownership. First, landownership involves the right to use the land for grazing animals, cultivating crops or other agricultural activities, what we call a *use right*. Second, a land-owning household has the right to alter the land and transform it, as well as profit directly from its productive capacities, what we call a *control right*. Third, ownership rights may include the right to transfer the land (and the corresponding ownership rights associated to it) to a third party, in any capacity. This right may be broken up into transfers within the household (through, for example, inheritance), what we call a *bequest transfer right*, and transfers between households (through sales, leases or as collateral), what we call a *full transfer right*. Fourth, and finally, a household may have the most secure property right, that of a fully transferrable, societally recognized property title, securing all the previous rights in in order to provide protection from seizure of land asset by a third party (including the state), what we call *titled right*.

Having delineated these rights of land ownership, we move on to applying this framework to our results. We can use, as a first example, the use and control rights. We note that all landowning households in all of our surveys hold these rights, no matter the institutional set-up. A household with land in, say, Guatemala, just like a household with land in, say, China, enjoys the right to use the land they possess and profit from it. In this respect, our results are completely comparable across all countries under analysis. Stated differently, with respect to use and control right-ownership inequality, our inequality estimates are completely equivalent across regions. However, if we were to compare, for example, titled right-ownership inequality, our results, as they stand, are not immediately comparable.

This is because, while households in Guatemala might have access or even possess, regularly defined property titles, which permit the sale of a land asset, households in China, due to its idiosyncratic institutional set-up, do not¹⁴. For this reason, with this strictest property right, we are only able to compare a subsample of the countries and households.

The baseline definition of ownership in this paper – that is – the definition which covers all countries is one where households with land enjoy three rights: use, control, and bequest land rights. This implies that all households which report being land holders in our analysis fully control and profite from their land, as well as possess the right to pass said land down within the household.¹⁵ Consequently, with respect to this definition, which we call **private household** ownership, all the inequality shares reported are comparable across all countries and regions. We can conceptually understand some stricter definitions compared to this baseline definition. **Marketable private household** ownership refers to land ownership including all previous rights as well as existence of and legal right to sell, lease, or use as collateral. Lastly, **titled private household** ownership designates all previous rights, as well as the possession of a legally recognized land title protecting from all expropriation of land by third party. The coverage of countries and households for each definition can be found in Figure D1. In future versions of this paper, inequality shares will be estimated using the last two definitions discussed.

¹⁴For example, land in China has, for the around the last 70 years, been owned collectively, and administered by the state in a registration system denominated hukou. Farmers have long-term contracts that may be renewed and passed on within a household, but may not be sold to other farmers for profit. For example, see [Vendryes, 2010](#) for more details on particularities of land tenure in China.

¹⁵Vietnam, for example, whose land is legally owned by the state, confers to all land holders the rights to fully exploit land as well as freely transfer it ([Ravallion and Van de Walle, 2008](#)). For less clear cases, such as for African countries with varying institutional contexts, the households designated as land owners only include households stating land as “purchased” or “inherited”.

<i>Definition of Ownership (Including which land property rights)</i>	<i>Subsample considered</i>
Private Household (Benchmark Definition): • Use • Control • Bequest Transfer	All countries, all households reporting land ownership
Marketable Private Household: • Use • Control • Full Transfer	All countries <u>except</u> China and Vietnam. Coverage within certain African countries will decrease.
Titled Private Household: • Use • Control • Full Transfer • Titled	All countries <u>except</u> China and Vietnam. Coverage within all countries will decrease even more as we consider only households with full property titles.

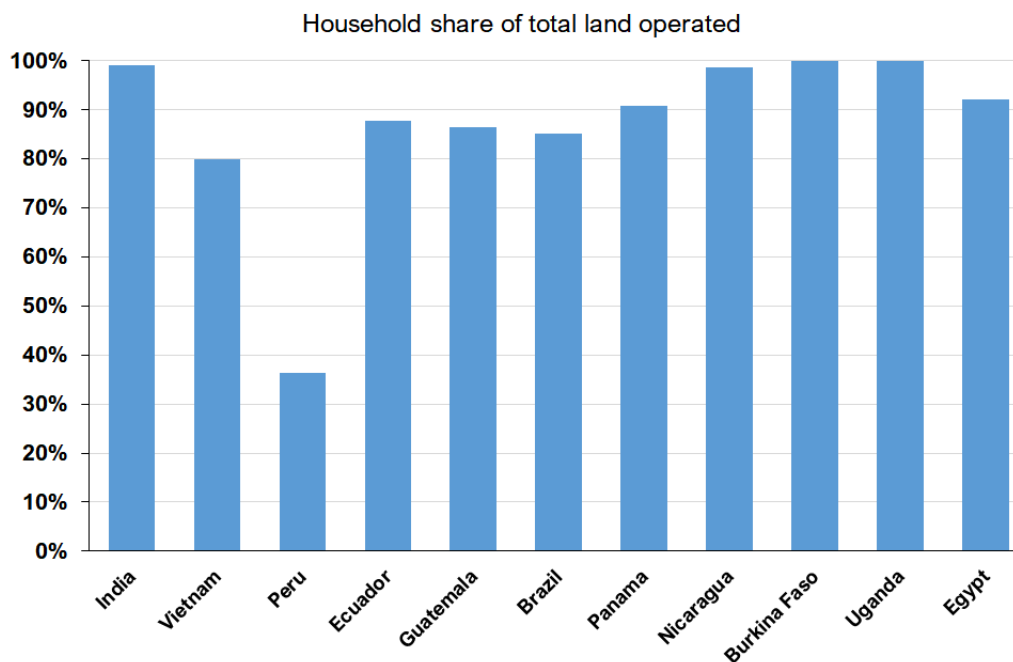
Fig. 7. **Figure D1:** Coverage according to different ownership definitions

Appendix E. Additional figures and tables

Region	Asia	Africa	South America and Caribe	Europe	North America and Oceania
Average	85%	89%	81%	53%	64%
Median	91%	89%	85%	57%	64%
Standard deviation	12%	7%	15%	19%	2%
Number observations	11	6	13	34	3

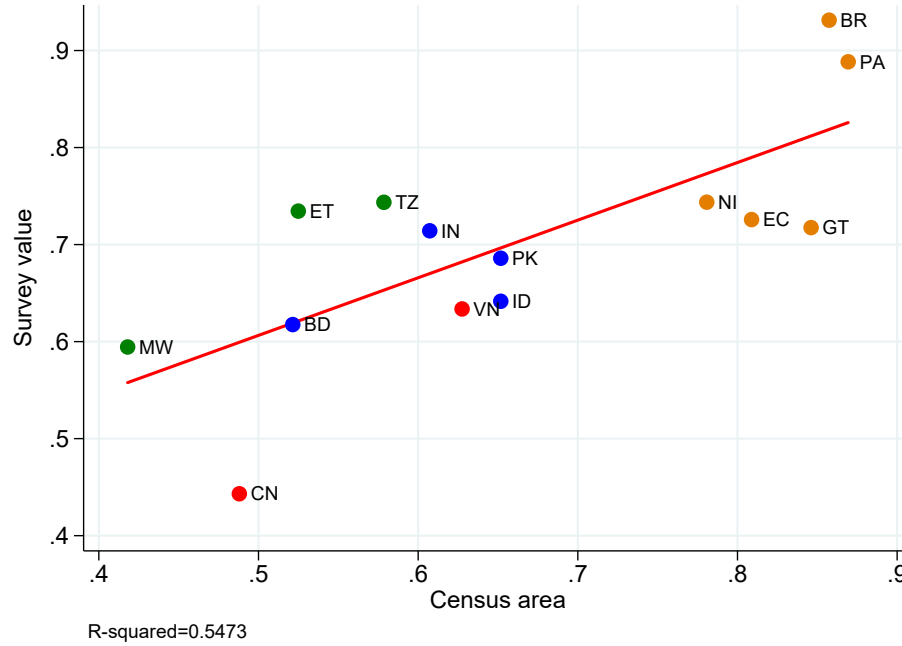
Note: This table shows summary statistics for the proportion of land operated that is also owned by the landholder across world regions. Source: FAO World Programme for the Census of Agriculture of 2010.

Fig. 8. **Table E1:** Share of operated land owned by the landholder



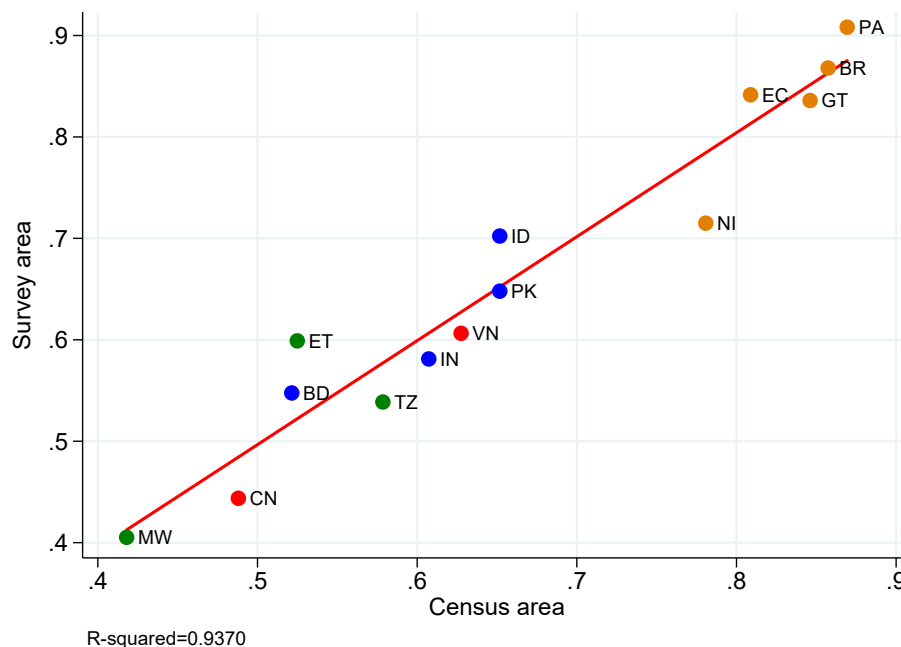
Note: This table shows the percentage of total agricultural land that is operated by households. Source: FAO World Programme for the Census of Agriculture of 2010.

Fig. 9. **Figure E1:** Share of agricultural land operated by households



The countries and the year of survey are as follows: BD – Bangladesh (2011); BR – Brazil (1996); CN – China (2002); EC – Ecuador (2014); ET – Ethiopia (2011); GT – Guatemala (2000); ID – Indonesia (2014); IN – India (2012); MW – Malawi (2010); NI – Nicaragua (2014); PA – Panama (2008); PK – Pakistan (2010); TZ – Tanzania (2018); VN – Vietnam (2014). For sources of data, see Appendix A.

Fig. 10. **Figure E2:** Gini index in selected countries: land area from census data vs. land value from survey data



Note: this figures compares the gini index for land area from census and survey data in the sample of countries included in figure appendix E2.

The countries and the year of survey are as follows: BD – Bangladesh (2011); BR – Brazil (1996); CN – China (2002); EC – Ecuador (2014); ET – Ethiopia (2011); GT – Guatemala (2000); ID – Indonesia (2014); IN – India (2012); MW – Malawi (2010); NI – Nicaragua (2014); PA – Panama (2008); PK – Pakistan (2010); TZ – Tanzania (2018); VN – Vietnam (2014). For sources of data, see Appendix A.

Fig. 11. **Figure E3:** Gini index in selected countries: land area from census data vs. land area from survey data (in countries for which land value from surveys is also available)